

6 A check digit is one method to validate a 5-digit number.

The check digit for a 5-digit number is calculated as follows:

- Each digit is multiplied by its position (the first digit on the left is in position 1).
- The result of each multiplication is totalled.
- $\text{MOD}(\text{Total}, 11)$ is used to calculate the check digit.
- If the result of the MOD function is 10, the check digit is 'X'.

For example:

The calculation of the check digit for the 5-digit number 30475 is as follows:

Multiply each digit by its position and total the results:

$$(3 * 1) + (0 * 2) + (4 * 3) + (7 * 4) + (5 * 5) = 68$$

Calculate the check digit with $\text{MOD}(\text{Total}, 11)$

$$\text{MOD}(68, 11) = 2$$

The check digit for 30475 is 2

This pseudocode algorithm should calculate and output the check digit of a 5-digit number:

```

01 DECLARE Total : INTEGER
02 DECLARE Digit : INTEGER
03 DECLARE CheckDigit : CHAR
04 Total ← 0
05 FOR Count ← 1 TO 10
06     OUTPUT "Please enter digit ", Count
07     INPUT Digit
08     Total ← Total + (Digit / Count)
09 NEXT Count
10 CheckDigit ← MOD(Total, 11)
11 IF CheckDigit = 10
12     THEN
13         OUTPUT "The check digit is X"
14     ELSE
15         OUTPUT "The check digit is ", Count
16 ENDIF

```

